

FGI-GSRx v2.1.1 — Release Notes

1. New Features in v2.1.1

1.1 Parallel Tracking Capability

FGI-GSRx v2.1.1 introduces support for executing tracking channels in parallel using MATLAB's **Parallel Computing Toolbox (PCT)**. Users can now significantly reduce processing time when handling long datasets.

Parallel tracking can be performed in **two alternative modes**:

Option 1 — Parallel Execution via Multiple MATLAB Instances

- The receiver automatically splits all tracking channels into **N single-channel jobs**.
 - It then generates a **batch script** containing all required execution parameters.
 - The user runs this batch script manually in the operating system command prompt.
 - Each MATLAB instance processes one tracking channel independently.
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Option 2 — Parallel Execution Using MATLAB PCT

- The receiver uses MATLAB's **PCT** to run all tracking channels in parallel.
 - The number of parallel workers equals the number of physical cores allocated to MATLAB.
 - After execution, the receiver automatically **combines all tracking output files** into a single dataset.
 - Acquisition, tracking, and Position-Velocity-Time (PVT) can now be produced in **one unified run**, together with the desired raw I/Q data.
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For further details and background methodology, please refer to:

Liaquat, M., Bhuiyan, M. Z. H., Islam, S., Pääkkönen, I., & Kaasalainen, S. (2024). An Enhanced FGI-GSRx Software-Defined Receiver for the Execution of Long Datasets. *Sensors*, 24(12), 4015. <https://doi.org/10.3390/s24124015>

1.2 New Configuration Parameters

To enable parallel tracking, users must configure the following parameters in their receiver configuration file (e.g., defaultReceiverConfiguration.txt, or any custom configuration file):

% Tracking option: parallel vs sequential

```
sys,parallelChannelTracking, true, % Enable parallel tracking mode  
sys,PCTenabled, true, % Enable MATLAB Parallel Computing Toolbox
```

2. Changes from v2.1.0 → v2.1.1

2.1 Added

\Folder\File	Remarks	Modification
\track\doTrackingPCT.m	Implements parallel tracking using MATLAB PCT	New function
\track\doTrackingSingleChannelParFor.m	Executes a single tracking channel using MATLAB parfor	New function

2.2 Modified

\Folder\File	Remarks	Modification
\documenteation\readme.txt	Updated readme.txt with a list of publications and information about obtaining raw IQ data files	
\main\gsrx.m	Added functionality to call doTrackingPCT.m	Modified
\track\CombineSingleTrackChannelData.m	Updated comments	Modified
\track\doTrackingSingleChannel.m	Updated comments	Modified
\track\initializeAndSplitTrackingPerChannel.m	Updated comments	Modified
\param\defaultReceiverConfiguration.txt	New parameter	Modified

	'PCTenabled' added	
\param\default_param_GPSL1_Chapter2.txt	New parameter 'PCTenabled' added	Modified
\param\default_param_GLONASSL1_Chapter3.txt	New parameter 'PCTenabled' added	Modified
\param\default_param_GalileoE1_Chapter4.txt	New parameter 'PCTenabled' added	Modified
\param\default_param_BeiDouB1_Chapter5.txt	New parameter 'PCTenabled' added	Modified
\param\default_param_NavICL5_Chapter6.txt	New parameter 'PCTenabled' added	Modified
\param\default_param_MultiGNSS_Chapter7.txt	New parameter 'PCTenabled' added	Modified
\param\default_param_GPSL1C.txt	New parameter 'PCTenabled' added	Modified
